Manual

Calculation of Bonus Wages/Incentive Wages

LLE-BPL 8.1

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Contents

1 LLE-BPL - Overview

Purpose

The *Calculating premium/ incentive wages* module is used to determine personal standard incentive wage components for calculating premiums. This is piecework wages that account for time wage components, overhead costs and average wage

Moreover, this module provides intervention options that make it possible to use bonuses to respond to unforeseen events, for example in the production process, and to exercise corrective influence over wage calculation.

The calculated results can be transferred via an interface to payroll accounting in the form of wage types.

Integration

This is based on the personal data derived from shop floor data entry. As an option, times derived from time and labor data management can also be factored in. Also, where applicable, the data can also be imported from and processed in external systems.

Features

The LLE-BPL Application Service (AS) offers numerous field-tested functions for illustrating the common incentive-based forms of payment:

- Personal calculation of standard incentive wage components, piecework, time wage, overhead costs etc., based on data entered in HYDRA BDE and PZE
- Accounting for incentive wage indicators in HR master data with more than one version
- Consideration of incentive wage indicators in machine and/or workplace master data
- Automatic recalculation of incentive wage when posted data is modified later
- The ability to define upper and lower limits used to generate warnings about the performance efficiency grades calculated
- Option to separate out time wage amounts from piecework processing depending on resource - performance accounts posted in BDE
- Control the required authorization for certain order types, e.g. for overhead cost time tickets
- Optional consideration of scrap quantities during piecework calculation

- Optional compensation for specified set-up times during piecework calculation
- Optional calculation of the daily performance efficiency rate based on attendance time derived from time and labor data management
- Optional entry of bonuses and deductions at the BDE terminal by employees with authorization methods
- Interface for transferring results to the payroll system in HYDRA standard format
- Extensive data protection and security functions using function authorizations to protect personal data by utilizing HR master data responsibility areas

2 Incentive Wage Calculation - Process

Basics

The HYDRA Incentive Wage Calculation uses the data that has been recorded by a BDE or PZW system. For the different systems of premium or incentive wage, the application uses different subsets of the shop floor data recorded to calculate the incentive wages.

- Postings of operations and persons recorded by the Shop Floor Data Collection BDE in combination with the posting of times to the different resource performance accounts (RPA) performed by the Machine Data Collection MDE.
- Approved bonuses and deductions recorded by the Premium and Incentive Wage LLE
- Changes of group recorded by the Premium and Incentive Wage LLE
- Attendance, absence and break times of the employees recorded by the Personnel Time Management PZW

In addition to the transaction data above, the master data stored in the system for orders and persons and the LLE payment rules affect the data processing for the calculation of wages. For example, a configuration in the master data releases specific persons and orders for the processing in the incentive wage and bonus calculation.

Another important component of the incentive wage calculation is the complete (time) data collection in the relevant HYDRA program module. The objective is to track the complete time between clock-in and clock-out of an employee and to post the times to the relevant operations in the HYDRA system. To fulfill this requirement, HYDRA provides the waiting period processing where all times that an employee is not logged on to an operation are automatically posted to a waiting period operation.

Multiple systems exist to calculate incentive wages. On one side there are systems to calculate person-related piecework, on the other there are purely group-related bonus systems. The premium system used specifies which data recorded in HYDRA is used to calculate the incentive wage.

Using data of the Shop Floor Data Collection (BDE)

The BDE personnel postings are often used to calculate individual piecework or person-related bonuses.

To calculate group bonuses, the BDE order postings and sometimes the BDE personnel postings specify the key figures used. To distribute group results to the separate persons, the system generally uses the BDE personnel postings.

In the BDE, the following data is automatically transferred from the operation to the BDE order and personnel posting when an OP or a person is logged on (condition: the HYDRA products to calculate premium and incentive wage and/or to calculate group bonuses are licensed):

- **Single piece specification t_e**
  Specified in format hours with 3 decimal places per 1000 pieces

- **Setup specification t_r**
  Specified in format hours with 3 decimal places

- **Single piece specification for the production resource t_{eb}**
  Specified in format hours with 3 decimal places per 1000 pieces

- **Setup specification for the production resource t_{rb}**
  Specified in format hours with 3 decimal places

- **Wage type**
  The wage type often specifies how a data record is used for the incentive wage calculation. Different wage types exist for piecework, bonus wage, time wage, overhead costs or other.

- **Premium group**
  If the HYDRA product to calculate group bonuses is licensed, then the premium group assigned to the machine is transferred to the order and personnel postings. Optionally and as part of a customization, the premium group can be entered on the BDE terminal when the order is logged on.

In rare cases of the incentive wage calculation, the operator position (function) of the person at the machine is relevant. If required, this operator position can be entered and then processed when the person is logged on. This is a custom function. Another special feature of the incentive wage calculation can be that the person can enter a premium indicator for the activity performed when the person logs on. This premium indicator is then used to calculate the formula-based premium wage. This functionality is implemented via customization.

Using data of the Personnel Time Management (PZW)

In seldom cases, the data of the Personnel Time Management (PZW) is used to calculate the incentive wage. You can use PZW wage type postings or the totaled work day result to calculate the piecework wage or an individual bonus.

In some cases, the PZW wage type postings are assigned to the premium groups via "Change of group". The PZW wage type postings are then used to calculate the key figures for the group bonus and/or to distribute the group results to the persons.

Using data of the Premium and Incentive Wage (LLE)

On the office client, you can optionally record bonuses and changes of group (additional license or customization) for the calculation of the Premium and Incentive Wage (LLE). It depends on the system used to calculate the incentive wage whether bonuses or changes of group are included in the calculation.

You can use bonuses to manually change the premium calculation.

You can use the changes of group to assign the person-related data records of the PZW or the BDE to premium groups.

Data processing

The postings of the employees for the orders they have processed are logged in the HYDRA system. On a daily basis, these postings are transferred to the data of the incentive wage calculation to calculate the wages. Because of the above mechanism, the incentive wage calculation only includes data up to the previous day because the attendance times of the current day are only available on the next day.

If you want to correct wage-relevant data, you directly use the editing functions of the shop floor data collection and the personnel time management and you therefore directly change the output data. The incentive wage calculation then processes the data with the next run of the wage calculation (usually on the next day). The data recorded in the HYDRA system, e.g. BDE personnel postings, is only stored in the system for a specified time (35 days by default). You can only make changes to this data during this time.

For the HYDRA Premium and Incentive Wage, you can configure a longer period for the retention of wage-relevant data on the HYDRA server. If you need a longer period of data retention, these settings can be changed in the system. If you are not sure about the configuration of the data management, contact MPDV.

Overview

The wage calculation is the core function of the Premium and Incentive Wage. The system uses the BDE and PZW postings of the employees to calculate the results for the time tickets or the work day results of the persons and premium groups.

There are two options to start the wage calculation:

1. Use the HYDRA scheduler to start a daily wage calculation that calculates all work day results for persons and premium groups up to the previous day.
2. Manually start the calculation via GUI for specified days and persons or premium groups. For more information, refer to the documentation of this application on the client.

You can repeat the wage calculations as often as you like. Make sure that the data of the relevant day is still available in the system.

For further information, refer to the following documents:

- Information on the settlement day.
- Information on the performance efficiency calculation.
- Information on the group bonus and premium areas.

Automatic recalculation of historical data

In mode *Required days until*, the wage calculation automatically recalculates all day results for persons and premium groups up to the date transferred. To this end, the system memorizes whether changes have been made to data of persons or premium groups. The relevant day results for persons and premium groups must then be recalculated. Also the automatic wage calculation entered in the scheduler starts in this mode and recalculates all relevant data up to the day before today.

The logs of the automatic wage calculations entered in the scheduler are recorded in the HYDRA system logs as application HYL_COMPUT. The manual wage calculation also creates entries here. The column *Number of data records* shows the sum total of the calculated day results for persons and premium groups. It provides an approximate evaluation of the volume that must be calculated.

The following activities trigger an automatic recalculation for the relevant days:

- Editing of bonuses (also approval/rejection) for the relevant person or premium group on the specified day
- Editing of PZW wage type postings (also approval/rejection) for the relevant person on the specified day (also if the data has been passed from external systems via interfaces)
- Recalculation of day results by the PZW labor time calculation for the relevant persons on the specified day
- Editing of BDE postings using the editing function for order-related postings of the relevant person or premium group on the specified day

Some editing activities do **not** trigger an automatic recalculation because very long periods of time or very large data volumes might be affected. In theses cases, the user must manually start a wage calculation in mode *Period* for the relevant persons or premium groups, if required. Examples of such activities:

- Editing of changes of groups (these can cover a very long period)
- Editing of master data of premium groups (a period cannot be limited effectively).
- Editing of assignments of premium groups to premium areas (a period cannot be limited effectively).

Combining consecutive time tickets

This function is used to combine BDE personnel postings that have been interrupted by an automatic shift change. This is required in case of an overtime after end of shift. With a manual recording of quantities, the problem is then that the first part of the posting has a duration, but no quantity. The second part of the posting then includes the complete quantity recorded with staff logoff, but includes only a short duration. If the wage calculation calculates each of these postings separately, then the resulting performance efficiency rates would be very low with posting 1 and very high with posting 2.

To avoid this problem, the wage calculation can combine similar BDE personnel postings that are directly one after the other in time and consider them as one posting and then create one time ticket.

To activate the function to combine consecutive time tickets, go to the *Basic settings for incentive wage*.

The following conditions must be fulfilled to combine two BDE personnel postings:

Column	Condition
Machine	must be the same
Person	must be the same
Order/operation	must be the same
Cost center	must be the same
Quantity units	must be the same
Times of postings	Logoff time of the first posting must be identical to the logon time of the second posting.
Wage specifications t_e, t_r, t_{eb}, t_{rb}	must be the same
Wage type	must be the same
Premium group •	must be the same
Operator position/function	must be the same
Premium indicator	must be the same
Performance units	must be the same
User fields	Numeric user fields must be 0 or empty with both postings. Alphanumeric user fields must be identical in the first and second data record.

When two BDE postings are combined, the data is either totaled or the data of the second posting is used:

Column	Action
Quantities	are totaled.
Quantity reasons	of the second posting are used. (The second posting includes the

	information on the event that triggers the logoff).
Times of postings	are identified using the logon time of the first posting and the logoff time of the second posting.
Shift information	of the second posting is used.
Durations and resource performance accounts	are totaled.
Interruption reason	of the second posting is used. (The second posting includes the information on the event that triggers the logoff).
Performances	are totaled.

3 Wage Calculation

Overview

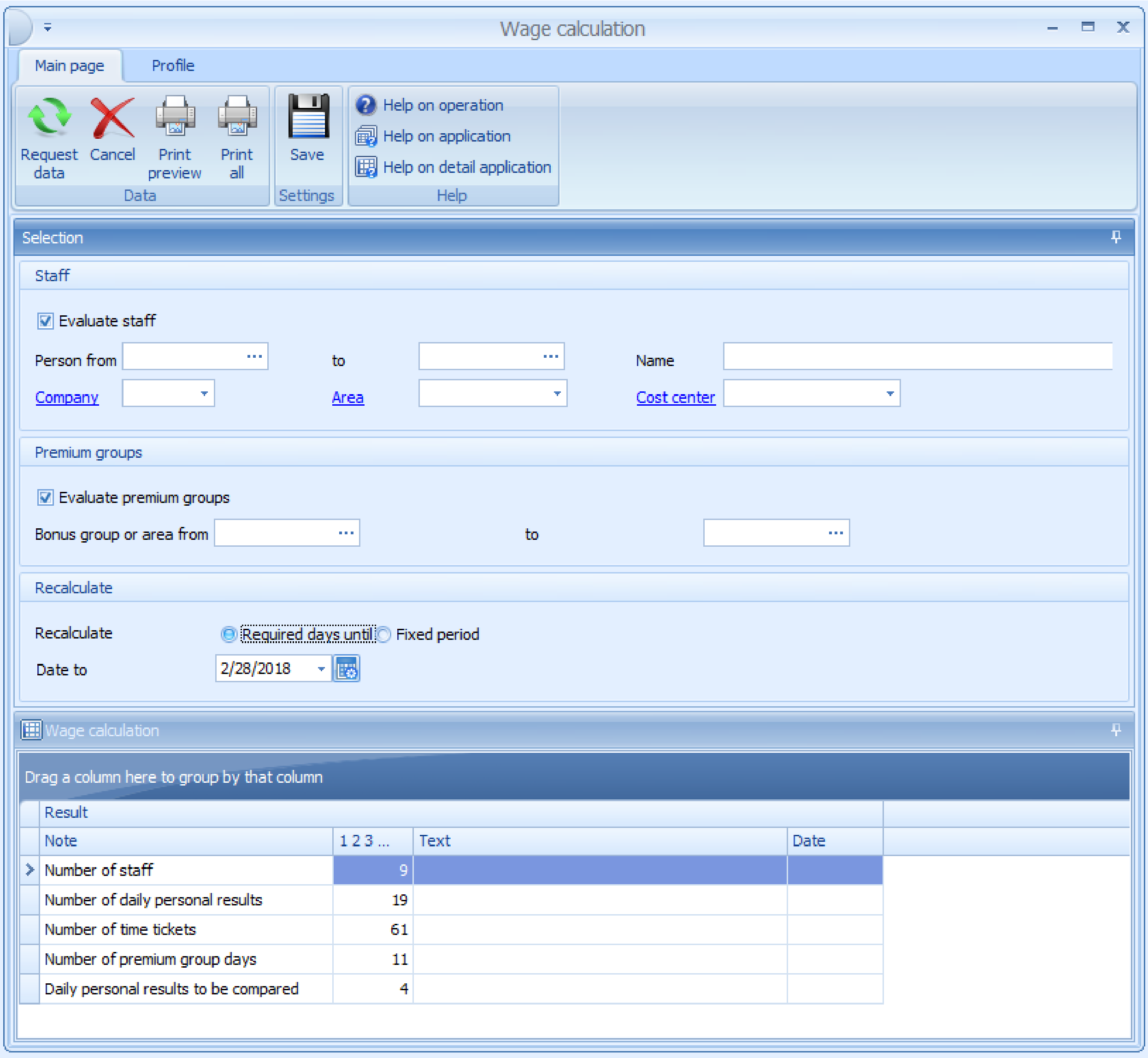
Menu	Human resources management → Incentive wage → Lohnberechnung
Transaction code	iwcalc
Function authorization	iwcalc.*

You use this application to start the wage calculation of the the incentive wage. The wage calculation uses postings of the employees (e.g. of the BDE and PZW) to calculate the time tickets and the work day results of persons and premium groups.

The Incentive Wage Calculation is described in a separate document.

The section below describes the manual start of the wage calculation.

In the course of the wage calculation, interrelationships can occur that are not predictable because persons can work for different premium groups at the same time. Therefore, you can hardly exclude or lock specific persons or premium groups when you start the wage calculation. Instead, the system does not allow to run different wage calculations at the same time. Only one user can start a wage calculation at a time. For the other user who starts a wage calculation, a relevant message is displayed in the list *Wage calculation*. This user must put off the start to a later point in time. This lock also affects the generation of the LLE interface file because it is not guaranteed that the data is complete when a wage calculation is run. In this case, the system does not output the usual wage calculation, but shows a warning message.



Selection criteria

The application provides the following selection criteria:

Evaluate staff

If this option is checked, the time tickets of the persons are recalculated. If this option is not active, only the time tickets of premium groups that have been recalculated are updated. You can limit the persons that must be recalculated using the other fields. If the selected persons worked in premium groups, it is possible that persons are recalculated who were not included in the original selection because these persons also worked in the premium groups involved.

Evaluate premium groups

If this option is enabled, the premium group results are recalculated. The user can use the other fields to further restrict the premium groups to be calculated. The system might also recalculate groups that were not included in the original selection because some of the selected persons worked in several premium groups.

Recalculate *Required days until*

All persons and premium groups selected for recalculation are recalculated up to the specified date. On the specified date, recalculation is also performed for persons and premium groups without existing wage calculation for the day. The activities that trigger an automatic recalculation of a person or premium group are described in the documentation Incentive Wage Calculation.

Recalculate *Fixed period*

A recalculation of all persons and premium groups is performed for the specified period. It is recommended to restrict the data using the personnel numbers and premium groups. For reasons of security, the system limits the period that can be recalculated to a maximum of 31 days in a row.

Field descriptions

Number of staff

Number of HR master data versions included in the original selection

Number of daily personal results

Number of recalculated personal work days

Number of time tickets

Number of recalculated personal time tickets

Number of premium group days

Number of recalculated days of premium groups

Note

Information on the completion of the wage calculation. The possible notes are described below.

Notes on the wage calculation

"Person 123456 day not computable. Reason: XYZ archiving."

With evaluation *Required days until*: For this person, the specified day must be evaluated, but the data required of the product group specified is no longer available. The earliest possible date that can be evaluated results from the last retention date according to the data management configurations of the specified product group. If the archiving program has not yet been performed for the data, the earliest possible date is identified using the earliest data records in the product group including a safety margin of one day. The PZW data must have been calculated by the workday evaluation.

"Locked by user XX, client XX, module XX."

"Locked. Please try again later."

Another user has already started a wage calculation or an automatic wage calculation is performed in the background. Only one wage calculation at a time can be performed in the system. Try again a short time later.

"Start date limited since data not available."

With evaluation *Fixed period*: The period specified starts before the earliest possible date that can be evaluated. *Date from* has been set to the earliest possible date that can be evaluated. See also note "Day not computable since data not available."

"ERR start date after end date."

With evaluation *Fixed period*: The *Date to* is before the *Date from*. This message can also be a subsequent failure of the message "Start date limited since data not available."

"ERR Maximum nbr of 31 days exceeded"

With evaluation *Fixed period*: You can select a period covering a maximum of 31 days. This limit is used to protect the user from accidentally entering incorrect dates in the date fields.

4 Incentive Wages Basic Settings

Overview

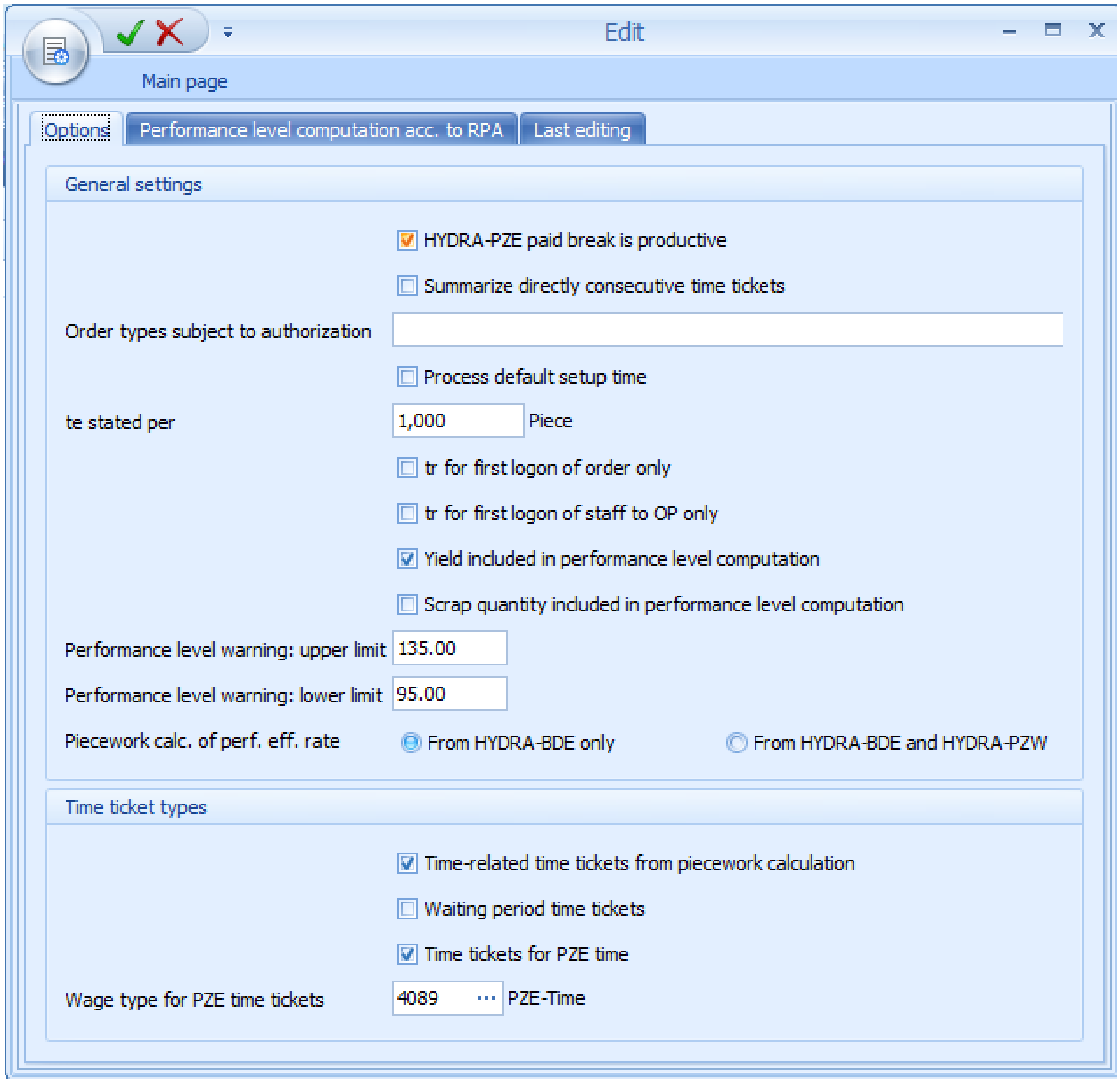
Menu	Master data → Incentive wages → Incentive wages basic settings
Transaction code	iwset
Function authorization	iwset.*

This application provides the basic settings for HYDRA incentive wage determination. When using the formula-based incentive payment LLE-FBL, the settings entered here cannot be overridden by customer-specific formulas.

Basic settings for incentive wage can only be edited. It is not possible to insert or delete data records.

Settings are divided into two sections.

The first page includes options for general settings and time ticket types:



Field Description for the "Options" Tab

HYDRA-PZE paid break is productive

If a performance level is calculated on basis of the PZE attendance time and the PZE time is not determined using the PZE wage type postings, this option can be used to control whether the paid breaks configured in the PZE are considered as productive time and are consequently included in the actual time for piecework calculation.

This setting is also considered in the *Labor time comparison*.

Summarize directly consecutive time tickets

If this option is activated, equal, directly consecutive ADE personnel postings are summarized to a single posting in wage calculation and consequently only one time ticket is generated. The ADE postings themselves remain unchanged.

Such postings mostly result from automatic shift changes.

A detailed description of this function can be found in the document on incentive wage calculation.

Order types subject to authorization

Here, ADE order types may be entered, separated by comma. ADE postings of these order types must first be authorized by foremen and/or specialists before they are included in the wage calculation.

Process default setup time

If this option is activated, the default time is calculated according to the following formula:

*Quantity m * Production time* t_e *+ Setup time* t_r.

If this option is not selected, tr is not considered and the default time is calculated according to the formula:

*Quantity m * Production time* t_e

Depending on the other options of 'Basic settings for incentive wage', quantity m is composed of yield and/or scrap.

t_e stated per x pcs.

Production time t_e is indicated for piece number x. By default, t_e is indicated for 1000 pieces in HYDRA. This value may only be modified in exceptional cases and in consultation with MPDV, as any change may have significant technical consequences.

t_r for first logon or order only

If this option is highlighted, the default values t_r und t_{rb} are only adopted in the ADE order posting (U or E record) upon the first logon of an operation in data collection. In the ADE posting records below, the fields remain empty.

t_r for first logon of staff to OP only

If this option is highlighted, the default values t_r und t_{rb} are only adopted in the ADE order posting (B record) upon the first logon of a person on an operation in data collection. In the ADE posting records below, the fields remain empty.

Yield included in performance level computation

If this option is active, the yield is included in the default time calculation.

Scrap quantity included in performance level computation

If this option is active, the scrap produced is included in the default time calculation.

Upper and lower limit for performance efficiency rate warning

If the limit values set here are exceeded or fallen short of in the calculation of a performance efficiency rate, an entry in the messages list is made by wage calculation. No entry is made in the messages list if both values are 0.0 or if the upper limit is less than the lower limit.

Piecework calculation of performance efficiency rate

This option determines how the piecework performance efficiency rate is calculated in the *daily results of persons*. The default time for the performance efficiency rate calculation in both options is determined using the ADE personnel postings on the basis of the quantity and single unit default t_e and, where appropriate, t_r. The *actual time*, however, is calculated differently according to the setting selected:

Only from ADE

The actual time results from the ADE personnel postings and, depending on other settings, according to the resource performance accounts or the entire labor time.

From ADE and PZE

The actual time for piecework calculation results from the PZE time of the employee on that day minus the time recorded on other time tickets.

$$Actual\ time = T_{pze} - T_{ZL} - T_{EA} - T_{GK} - T_{KAR} - T_{GRP}$$

This means the times of the time ticket types *Time wage, On-the-job training, Overhead costs, Waiting period and Group incentives* are deducted from the PZE time. The remaining time is the actual time for piecework calculation in the daily results of the employees.

The PZE time T_{pze} is determined in accordance with the following rule:

1) If there is a wage type with an active *Labor time for incentive wage* identification, the PZE time is composed of the PZE wage type postings and represents the total of all wage types with active *Labor time for incentive wage* identification.

2) If there is no wage type with an active *Labor time for incentive wage* identification, the total attendance time of the daily result calculated by Personnel time management is considered as PZE time.

Time-related time tickets from piecework calculation

If this option is activated, specific resource performance accounts can be eliminated from the piecework time for BDE postings to piecework-capable operations and, where required, be classified with a wage type for time wage. Please also refer to the further descriptions of LLE basic settings.

Waiting period time tickets

If this option is activated, time tickets are created from BDE waiting period postings. This requires use of waiting period processing of BDE. Waiting period time tickets are assigned with a wage type by standard wage type determination according to the waiting period order for the employee waiting period.

In incentive payment determination, waiting period postings may only be compensated for as time or overhead costs wage in individual incentives, since there is no relation to an incentive-relevant time. As this does not constitute appropriate processing in incentive payment determination, waiting period postings should be avoided at the order entry stage. This is supported by a good posting discipline of employees. Waiting period postings which do appear should subsequently be corrected to appropriate activities by maintaining order-related postings.

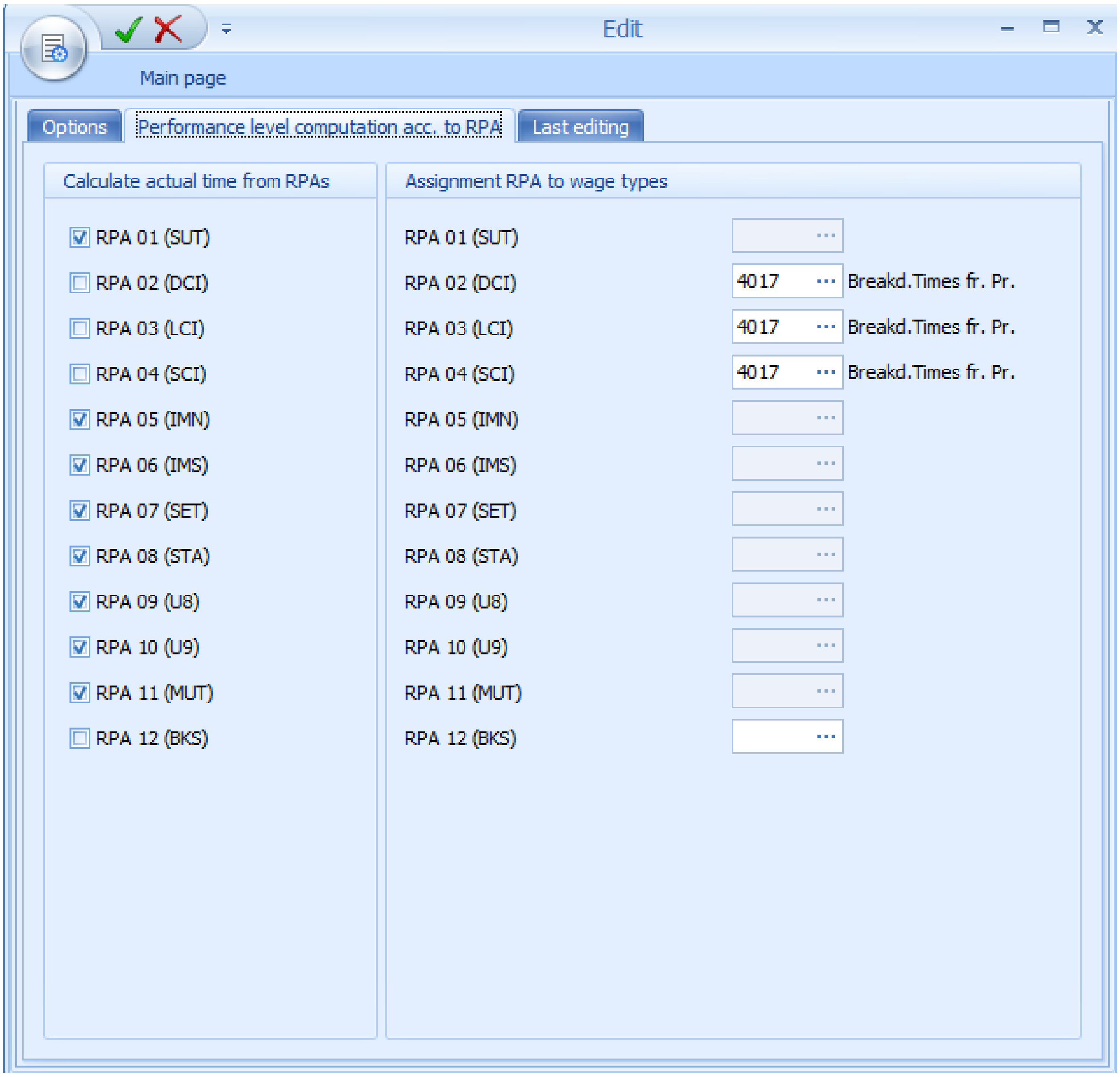
As regards group incentives, waiting period postings usually cannot be reasonably allocated to an incentive group and should therefore be avoided.

Time tickets for PZE time, PZE wage type

If this option is activated, a PZE time ticket is created for the employee's entire PZE attendance time per day. The time ticket includes the wage type according to the basic settings.

The second page of the LLE basic settings includes options to control the actual time determination for piecework wage and time-related wage portions on the basis of resource performance accounts:

Field Description for the "Performance level computation acc. to RPA" Tab



Calculate actual time from RPAs, RPA 1 through RPA 12

If these options are selected, only the selected resource performance accounts RPA 1...12 are used for the actual time calculation. If no resource performance account is activated, the entire proportional labor time is used.

Assignment RPA to wage types, wage type for RPA 1 through 12

RPAs can be allocated to wage types which result in time-related tickets and are hence considered as time wage times in the confirmation to a superordinate wage system. A resource performance account can only be allocated to a wage type if it is not used to determine the piecework actual time.

5 Performance Level (LLE)

Synonyms

The term "Performance efficiency rate" is commonly used in the context of the piecework wage calculation.

Definition

The performance level is the measurement of performance used to calculate premium/incentive wages. It is the results from a target/actual comparison and is shown as a percentage.

Most of the time, you compare a target time with the actual time. The target time is usually specified in such a way that it doesn‘t exceeded by the actual time, so that the performance levels are normally above 100%. This is due to improve employee motivation. To say "I achieve more than 100%" is more motivating than saying "I have only achieved 90% of the target performance".

Performance level for piecework

The performance level is calculated in the classic piecework rate by putting the target time in relation to the actual time. Data for piecework calculation origin from the BDE personnel postings (B records). The "Wage calculation" generates from the BDE personnel postings valuated time tickets which include the performance efficiency rate.

$$performanceLevel = \frac{standardTime}{actualTime} \cdot 100\%$$

- The *actual time* in piecework depends on the basic settings for incentive wages. However, this is usually the proportionately calculated working time deduced from the logon of a person in the order data entry (B records).This is calculated proportionally for multiple machine operation. You can also use the basic settings for incentive wages to specify that the *actual time* for piecework is only made up of certain RPAs used for personnel postings.

- The target time column results from the single piece target t_e and the quantity as well as the setup time target t_r stored at the operation:

- $standardTime = \frac{quantitiy \cdot t_e}{1000} + t_r + bonuses$

- The t_e is usually listed in HYDRA in hours per thousand pieces [h/1000].

- The quantity is made up of the yield or yield + scrap, depending on the basic settings for incentive wages.

- The t_r is only offset if this is activated in the basic settings for *incentive wages*.

- The collected bonuses are added to each time ticket that matches the employee's personnel and order number on the date and, as a result, is totaled in the person's daily result.

The performance level is calculated for piecework time tickets and in the daily results of the persons.

A performance level is also calculated for time tickets with other time types if the required basic data is available. However, these performance levels only serve as information for the particular time tickets and are not included in further payroll accounting.

To correctly determine the target time for single piecework, the person-related collection of the produced quantities in the order data collection is a mandatory requirement. A quantity posting to the logged on persons takes place in HYDRA for the following terminal postings:

1. Log off a person from an order/workplace

2. Partial confirmations/uploads

3. Interrupting and completing orders:
   1) at group workplaces
   2) at individual workplaces: for the person who has been logged on for the longest time, but only if the option "Quantity posting to person" is activated when configuring the machine or workplace.
   If one of the two conditions is not fulfilled when interrupting or completing orders, the quantities are only posted to the operation, and it is not possible to determine the correct performance level for machine operators!

When using "Formula-based premium/ incentive wages", customer-specific formulas can result in different calculation methods for performance levels.

Performance level for group incentives

You can also calculate performance levels for group incentives. The Calculation rules depend on the type of group incentive.

6 Evaluation Date

Definition

The settlement date defines to which logical settlement date a data record belongs. In particular in night shifts the settlement date may be different from the posting date.

Synonyms

Settlement date, settlement day.

Example

A night shift starts on 10:00 pm and ends on 6:00 am of the next day. An order posting or a clocking record from personnel time management from 4:00 am to 6:00 am of the following day will, however, still belong to that day on which the night shift starts. The settlement date will be that day, on which the shift starts.

Assignment of the evaluation date of postings from shop floor data collection for incentive wage and labor time comparison

Both the labor time comparison and the incentive wage calculation depend on a uniform assignment of the time and labor data from time and attendance and of the recorded postings from the shop floor data collection to one consistent settlement day. In particular in case of night shifts and irregular working times this is a certain challenge since the definition of the settlement date in PZW and BDE itself is not carried out according to exactly the same rules. In general, the assignment of the working time of a person to a settlement date from PZW is leading since the subsequent day calculation carried out by PZW allows for a coherent assessment of a work day, which is not possible in the online recording of BDE. In BDE, a personnel posting is saved online without that it is known whether there will be additional personnel postings for the same work day or not.

If therefore day-wide shift models with a night shift possibility are defined in the PZW or BDE, this may lead to differences between PZE and BDE in the labor time comparison since BDE postings are assigned to a different day than the corresponding PZW time due to varying shift models and for evaluation purposes.

Please check in those cases the BDE postings also for the adjacent days in the maintenance of postings dialog. To assign the BDE personnel postings to a settlement date, two columns are used that can be displayed in the list of the maintenance of postings dialog:

Field settl. date

If the settlement date is completed, the BDE-posting will be assigned to that day. The field will be completed by the PZW labor time calculation or the editing of postings relating to orders or the wage calculation.

This means that the field for all postings of the current day will normally first be empty and only be updated on the next morning through the PZE labor time calculation and/or the wage calculation. This means that postings of the current day may initially be incorrectly assigned to the day before. This incorrect assignment will in general automatically disappear on the next day.

Please note: The PZW labor time calculation only fills out the settlement date in the personnel postings of BDE if one of the functions LLE-BPL (calculation of bonus/incentive wages), SIS-APB (comparison of labor/shop floor times) or SIS-NPB (subsequent input of labor/shop floor postings) has been licensed.

Field shift date

If the field *Settl.date* is empty, the field *shift date* will be used for assignment purposes. If the shift date does regularly not match the PZE assignment, the BDE shift models must be adapted.

Assignment by labor time calculation (personnel time management)

By default, the assignment of the BDE postings to PZE work days is made according to the following rules. Please note that these default rules may have been changed by the customizing of the incentive wage determination.

BDE personnel postings (B records)

Personnel postings are assigned to the PZE personal performance (day) if they reach into the rounded working times of the posting person in a time frame of +/- 2 hours. If a personnel posting takes several days, it will be assigned to the first possible PZW personal performance (day) since the log-off has usually been forgotten in those cases.

Order related BDE postings

Order-related BDE postings may be accounted for in the calculation of a LLE group incentive. For the BDE/PZE comparison itself they are not important.

BDE order postings (U/E/T records) are assigned to the PZE personal performance (day) of the posting person to the extent that their log-off time lies in a time frame of +/- 2 hours to the rounded working times of the posting person. If a BDE order posting takes several days, it will be assigned to the last possible PZE personal performance (day) since the posting person of the BDE order posting will be defined from the log-off event and since the person, who logs off, posts the quantities. If no personnel number is recorded in the BDE order posting, the settlement date cannot be assigned on the basis of the time and attendance function.

Assignment by wage calculation (incentive wages)

Even if Personnel Time Management (PZW) is not used in HYDRA, wage calculation as part of incentive wage determination results in BDE personnel postings being assigned to a settlement day that might deviate from the BDE shift date. This is due to the fact that a person's working day is to be considered as completed even though the person works from night shift until the next day's early shift.

Please note that these default rules may have been changed by the customizing of the incentive wage determination.

BDE personnel postings (B records)

By default, the assignment is made according to the following rules:

1) If a BDE personnel posting starts between 10.00 a.m. and 11.00 p.m. it pertains in any case to the date of logging in.
2) All BDE personnel postings starting after 11.00 p.m. and not starting later than 2.00 hours after the previous BDE personnel posting are still assigned the considered settlement day, provided they do not start later than 10.00 a.m. of the next day.
3) If a BDE personnel posting ends after 11.00 p.m. and has a gross duration of more than 12 hours, this posting will not be affected by this rule. This is reasonable as it avoids unintentional summarization of working days over night if persons forgot to log off.

If HYDRA Personnel Time Management (PZW) is in use, the end of day limit (11.00 p.m.) is taken dynamically from the PZW end time of the affected daily personnel performance. The end of day limit is defined based on the PZW end time. Optionally, this limit can be extended by a configurable tolerance by MPDV customizing services.

The start of day limit (10.00 a.m.), end of day limit (11.00 p.m.) and the maximum gap (2.00 hours) can be modified by MPDV customizing, if required, to be able to assign the night shift to the next day, for example (start of day limit 2.00 a.m., end of day limit 3.00 p.m.).

BDE postings relating to orders

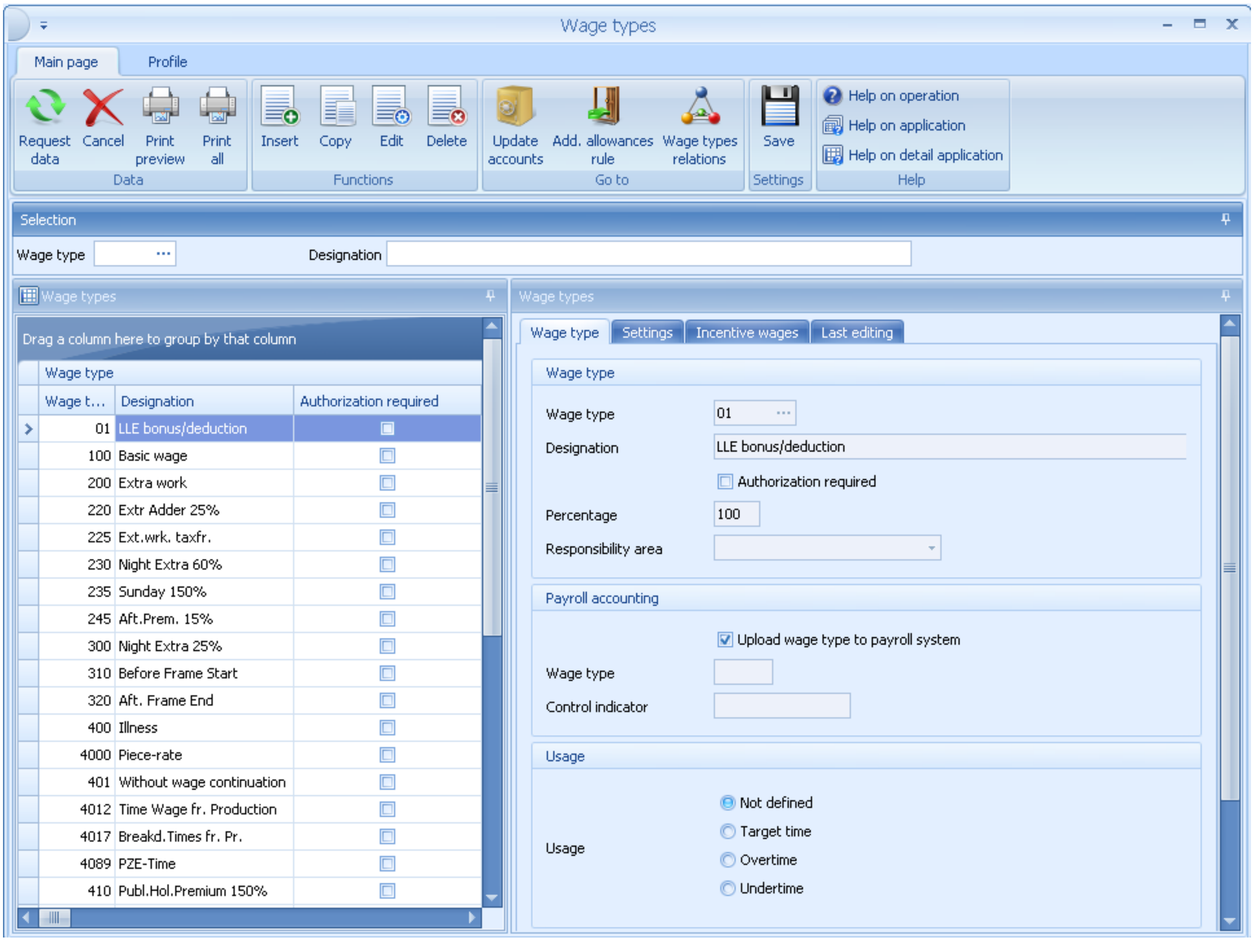
Order-related BDE postings might be included in the calculation of an LLE group bonus. In this case, wage calculation does not assign them separately to a settlement day but takes over the assignment made in BDE (shop floor data collection) or PZW (personnel time management).

7 Wage types

Overview

Menu	Master data → Labor time → Wage types
Transaction code	waty
Function authorization	waty

Wage types are different categories to group times with different information (e.g. night shift, overtime, etc.). We distinguish between basic wage types that are used for the payment of special working time and bonus wage. Usually, different types of absences are also specified as different wage types.



Field description tab "Wage type"

Wage type, name

Alpha numeric identification of the wage type and name.

Authorization required

This option is used to define a wage type with required authorization. If the option is not active, the system requires authorization somewhere else (e.g. in the payment day type).

Percentage

The percentage with which the wage type is compensated. Specifying a percentage only has an effect if the wage type is to be posted to an account. Otherwise, this is a comment field.

Entries with 0% are not posted.

Responsibility area

A user is only authorized to edit this wage type if he or she has authorization for the assigned area of responsibility.

Confirm wage type to payroll system

If an interface to Payroll Accounting exists, you can use this option to specify whether or not this wage type is transferred to the interface file.

Payroll wage type

You use the wage type to post information to the payroll department. This field is not processed in all interfaces.

Payroll control option

A field for customer specific processing.

Purpose

Specifies whether the wage type should be used to calculate planned working time, overtime or undertime. It is also possible not to specify anything. The wage types marked with Overtime are listed in the Overtime column of the time sheet. The same applies to the use of Undertime, but these times are displayed as negative.

Type

Specifies whether the wage type is a basic or a bonus wage type.

Field description tab "Settings"

Processing

Note on how this wage type is used. This is a comment field and can be left empty.

Selection field

A field for customer specific processing.

Average Type

The field "Average type" is processed with the aid of interfaces to transfer "Monthly wage types" to the payroll systems LOGA and Abacus. You can find further information in the description of the interfaces.

Rounding of wage type

The fields "Interval" and "Limit" (both in the format hours:minutes) can be used for rounding the daily duration of a wage type. The interval forms the points in time used to round up or down. The limit specifies up to what point in time the system rounds down during the interval and when it rounds up. If no rounding to wage types is required, you do not need to make an entry.

Wage types are rounded after the "Additional allowances rule" and the "Wage type interaction" were processed.

Use wage type for comparison with BDE.

You can use the Comparison function to compare data in the order data entry for rounding in personnel time recording. This is done with the wage types that are marked here.

Delete wage type after comparison with BDE.

If this wage type is only a processing wage type for comparison and can be deleted after the comparison.

Field description Tab "Incentive wage"

Time type

This field specifies the "Time wage, "Piecework" and "Overhead costs".

Labor time for incentive wage

This wage type is used to deduce the PZE labor time from the PZE wage type posting when you calculate the performance efficiency rate for piecework from ADE and PZE.

If the wage type is activated with this option, then the PZE labor time is always deduced using the PZE wage type posting no matter what person. If the wage type cannot be activated with this option, then you use the attendance time from the PZE as the labor time.

Incentive wages option

You only use this option if you calculate a formula-based incentive wage with a customized processing.

Labor time for group bonus

This field controls how PZE wage type postings are included in the calculation of the group bonus using formula-based incentive wages. This field is not relevant if you have a standard group bonus without formula-based incentive wage calculation.

Not included in the group bonus

PZE wage type postings for this wage type are not included as labor time in the group bonus.

Using cost center for posting

The cost center in the PZE wage type posting is interpreted as a premium group. In this case, the cost center for the PZE and the premium groups of LLE must be identical. Transfers to other premium groups can be achieved by manually assigning:

cost centers in the PZE clockings and postings

temporary cost centers

HR master data versions

cost center entries at the PZE terminal

cost center changes.

Using premier groups from the HR master data

With this option you assign the PZE wage type postings entered in the premium groups using the premium group of the HR master data. Persons can be transferred to other premium groups on a daily basis by creating HR master data versions.

Using group assignments

You use the function "Change of group" to assign people to the premium groups down to the exact minute. The assignment from the group changes is transferred to the PZE wage type postings for this wage type and then you can include the wage type posting for the group calculation. In order to do so, you separate the wage type postings if a group change takes place during the posting.

Quantity determination by

You use this option to control how the quantities for piecework are calculated when persons are posted in the ADE. This is relevant for wage type with the time type "Piecework".

Basic settings

LLE basic settings The system calculates the quantities for the time ticket, which includes scrap and yield from the primary quantities if the setting is made.

Wage type

You can use the matrix to set which quantity fields of the ADE posting are used to calculate the quantity for the time ticket.

Toolbar

Update accounts

Update accounts With "Update accounts" you specify which wage types are used to increase or decrease amounts for certain accounts.

Additional allowances rule

You use the option "Add. allowances rule" to post an additional bonuses if employees work on special days. Additional allowances rule Likewise, fixed special payments such as fare, lunch money or similar can be made.

Wage types relations

You can configure interactions between wage types Wage type interactions .

8 Time Type

Definition

The time type classifies the time tickets in the premium/ incentive wage. By default, the time type is used to control the calculations for time tickets.

The master data of the wage types and the LLE basic settings can be used to define which time times are to be created by HYDRA and how the time tickets will then be calculated.

"Piecework" time type

The piecework time tickets serve as basis for individual piecework. Only for piecework time tickets there is a calculation of the performance efficiency rate based on standard time, bonuses and reductions and actual time.

Depending on the wage type and the basic settings, the yield, scrap and set-up standard time t_r of the personnel postings (B-records) will be taken into account in the calculation of the standard time.

Granted bonuses will be noted in a separate column in the time ticket and be transferred to the standard time when the performance efficiency rate is calculated.

By default, piecework time tickets will be accounted for in the daily personal performance with duration, standard time and bonuses. This is how they define the performance level of the daily personal performance.

Time type "bonus"

The granted bonuses are also available as time ticket. As time type for these time tickets the smallest wage type with the time type "bonus" ZUS is used. If there is no such wage type, the wage type "01" is assigned.

Time type "Time wage"

Time-related time tickets are used to pay for production orders that are not piecework-relevant by an hourly rate.

Time-related time tickets from piecework orders

These time-related time tickets are used to pay for non-piecework-relevant times orders from piecework-relevant production orders by an hourly rate. A time ticket will be generated for each non-piecework-relevant resource performance account. Also for overhead cost orders such time tickets will be generated to the extent that the processing of overhead cost order time tickets is activated in the LLE basic settings.

The assignment of resource performance accounts to time wages from piecework is made in the incentive wage basic settings.

"On-the-job training" time type

Time tickets for on-the-job-training are used to pay for specially identified employees per hourly rate. As regards their compensation scheme they are identical to the time wage and have their own time type only for evaluation purposes.

"Overhead costs" time type

Time tickets for overhead costs are used to pay for overhead cost orders as time wage. As regards their compensation scheme they are identical to the time wage and have their own time type only for evaluation purposes.

Depending on the order type, it must be specified in the wage type determination that a wage type of the time type "overhead costs" will be used. By default, these are the order types 1 (GK) or 4 (GK II).

"Waiting period" time type

Time tickets for waiting periods are used to pay for ADE waiting periods as time wage. As regards their compensation scheme they are identical to the time wage and have their own time type only for evaluation purposes. Waiting period postings result from the shop floor data collection when waiting period processing is activated in the basic settings and the allowed waiting periods between postings have been exceeded.

When PZE time and attendance is used in HYDRA, the determination of the performance efficiency rate related to PZE should be preferred to waiting period postings.

"Attendance time" time type

Time tickets with the time type "attendance times" will be determined from the attendance time of the HYDRA personnel time management. They are only used to represent the PZE time in the HYDRA incentive wage and are taken into account as duration in the daily personal performance if the basic settings for the incentive wage are made correspondingly and will thus define the performance level in the performance level computation in personnel time management.

9 Interface to Payroll Accounting

Overview

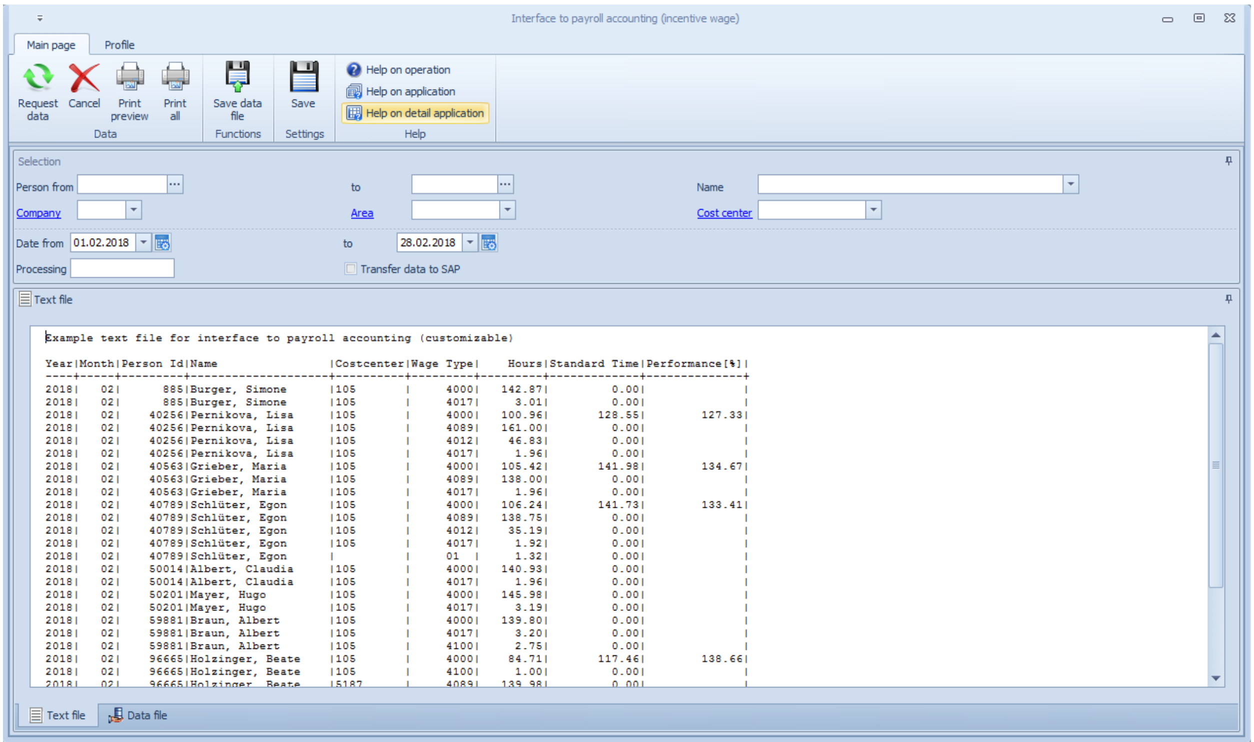
Menu	Human Resources Management → Incentive Wage → Interface to Payroll Accounting
Transaction code	iwipr
Function authorization	iwipr.*

The uploads to the payroll accounting are not performed automatically. The uploads are performed manually. For the upload, the time sheets and the bonuses of all employees are provided in an interface file on the HYDRA server in the HYDRA directory. This interface file covers a period of time that you are free to specify. A new file (hylrueck.dat) is created each time you call the upload function. You can save the file under any name on a data medium using a function key on the HYDRA console. For information on the data record structure of the upload file, refer to the section "Upload of wages" in the HYDRA documentation "Interface to payroll accounting".

If you use the "incentive wage based on formulas", you can define the contents and formats of the interface using custom formulas and scripts that are different to the ones shown in this document.

Note:
While the wage calculation is running, some wage data is not available for other evaluations. For this reason, you cannot create the LLE interface file and run the wage calculation at the same time. A locking mechanism is used to ensure this. In this case, the system does not show the usual screen of the interface file, but a respective warning.

When you have started the evaluation, the interface file is displayed.



Selection criteria

Date from / to

You can create the interface for single days, if required. But usually, the upload is performed for calendar months. The system populates the date fields with beginning and end of the previous calendar month.

If required, the system automatically archives the incentive wage data (time tickets and results of premium groups) in the long-term data area. If you process other data with the "incentive wage based on formulas", e.g. data of the personnel time management, you must ensure via customization that the data is stored for a sufficient period of time.

Transfer data to SAP

If an additional function for the direct upload of data to SAP-HR is active and if this option is enabled, the data is directly uploaded to SAP (update run). If this option is disabled, the data is only displayed on the screen (test run). You can make as many test runs as required before finally upload the data to SAP.

Identification and customization of the source system SOURCE_SYS

Many upload interfaces to SAP include the target SAP system as SOURCE_SYS in the data passed.

When the HR master data has been downloaded in SAP format to HYDRA using the HR-PDC, SAP also transfers the source system of the person. This system is stored in the sixteenth freely configurable info field of the HYDRA HR master data. No further configuration is required.

If the HR master data is maintained in a different way, this entry might not exist. You can then identify the source system for the upload using the ALE configuration in HYDRA (ALE = Application Link Enabling). To this end, the source system of an active logical SAP system is read. You can set this system in HYDRA via INI configuration.

You identify the source system using the following rule and priority. If a source system could be identified using the listed rules in the specified order, then the other rules are not executed.

1) Entry in info field 16 of the HR master data

If an entry is available in this field, this entry is interpreted as source system for the upload.

2) Via logical system from INI configuration for personnel number

Via INI configuration, a logical SAP system is specified for the personnel number:

Name of INI	"HR-LOGSYS"
Section	*required logical system*
Key	"PNR"
Value	Personnel number of the required person.

The active source system of the logical system is then identified.

3) Via logical system from INI configuration for the company

Via INI configuration, a logical SAP system is specified for the company defined in the HR master data:

Name of INI	"HR-LOGSYS"
Section	*required logical system*
Key	"FIR"
Value	Company.

The active source system of the logical system is then identified.

4) Via logical system from INI configuration, default entry

Via INI configuration, you can make an entry to generally specify a logical SAP system:

Name of INI	"HR-LOGSYS"
Section	*required logical system*
Key	"ALL"
Value	"Y"

The active source system of the logical system is then identified.

5) Default identification

The active source system of the logical system "SAP" is identified.

If no source system could be identified using the listed rules, the field remains empty.

Detail applications

You can switch between the display of the text file and the data file. By default, the display is identical. If you use the "incentive wage based on formulas", you can display readable information in the text file via customization by a specialist.

10 Premium/ Incentive Wage Uploads

Wage types are transferred to the payroll or HR information system in the **hylrck.dat** file.

10.1 Data record structure for incentive wage uploads

The interface structure matches that of the PZE interface file except that some of its unused fields are filled in with specific incentive wage data.

Therefore, the information in the **Data type** column has the same meaning as in the PZE interface file, so we will not explain it again here.

This file is structured as follows:

Field/ meaning	Position	Data type
Record type	1	always "760"
Company	4	C3
Area	7	C8
Settlement year	15	N4
Settlement month	19	N2
Settlement number	21	C1 = EMPTY
Personnel number (left justified, filled with EMPTY)	22	C8
Last evaluation day	30	N2
Wage type (left justified, filled with EMPTY)	32	C4
Preceding sign for wage type hours	36	C1 = +
Hours for wage type	37	N5.2
Full days absent	42	N3 = 000
Partial days absent	45	N3 = 000
Different wage group	48	C3 = EMPTY
Different hourly rate	51	N5 = 0
Amount	56	N7 = 0
Year of supplementary payment	63	N4 = Empty
Month of supplementary payment	67	N2 = Empty
Executing (master) cost center	69	C8
Charged cost center	77	C8
Order number	85	C10 = EMPTY
Work sequence	95	C4 = EMPTY
Comments	99	C18 = EMPTY
Premium group	117	C10

Performance efficiency rate	127	N6.3
Reserved for other incentive wage data	133	C9 = EMPTY
Document number	142	C5 = EMPTY
Posting indicator	147	C1 = "1"

10.2 Description of data fields for incentive wage uploads

Company:

The company from HR master data is entered here.

Area:

The area from HR master data is entered here.

Last evaluation day:

The last day of the date selection, e.g. "30" or "31"

Wage type:

Wage type from the personal time tickets

Preceding sign for wage type hour:

Is always "+".

Wage type hours:

Time that is to be posted to the wage type entered in the data record. The two decimal places are stored in industrial minutes.

Full days absent:

Filled in with 000.

Partial days absent:

Filled in with 000.

Year, month of supplementary payment:

Empty.

Executing cost center:

Person's master cost center.

Charged cost center:

Cost center from time tickets. The wage type sum is transferred separately to cost centers.

Order number, work sequence

Unused in the default.

Premium group

Premium group from time ticket. The sum total on the wage types is transferred separately to premium groups.

Performance efficiency rate

The total performance efficiency rate shown on piece-work time tickets for the settlement period is transferred in this field using three decimal places. A performance efficiency rate of 131.234% is converted to 131234 in this field. Six zeros (000000) are transferred on non-piece work time tickets.

10.3 Example file:

```
1        10        20        30        40        50        60        70        80        90        100       110       120       130       140    147
+--------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+-------
760BSPLLE     200802 40563   294000+11531000000   000000000000     105     105                                                  122907             1
760BSPLLE     200802 40563   294017+00139000000   000000000000     105     105                                                  000000             1
760BSPLLE     200802 40789   2901   +00464000000   000000000000     105                                                          000000             1
760BSPLLE     200802 40789   294000+11613000000   000000000000     105     105                                                  131619             1
760BSPLLE     200802 40789   294012+03519000000   000000000000     105     105                                                  000000             1
760BSPLLE     200802 40789   294017+00133000000   000000000000     105     105                                                  000000             1
```